

THE STARSHIP GALILEO

DEEP SPACE TRAVEL AND COLONIZATION

INSTRUCTIONS MANUAL

THE GEMINI PROTOCOLS

PHASE 165

THE QUINN TETHER GAUNTLETS

The Spider-Man rule, Hollywood struck
brace, then fire



the thrown anchor

KEVLAR · CO2 · NIMH REWIND

BRACE BEFORE FIRING

Crew system — v1.0 in force

GP-IM-165

Revision v1.0 — 23 August 2026

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VETTED BATCH 20 — TETHER PASS · BENCH LIST OWED

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LICENSE: CERN OPEN HARDWARE LICENCE V2 (CERN-OHL-W-2.0)

THE GEMINI PROTOCOLS — CHAPTER 165

PHASE 165: THE QUINN TETHER GAUNTLETS — WRIST-MOUNTED LINE THROWERS — STATUS: CURRENT — TETHER MECHANISM PASS; ORDINARY BENCH TESTS OWED. The Spider-Man rule, with the Hollywood physics struck out. Phase 165 mounts the line-thrower on the wrist: a short arm gauntlet firing thin Kevlar line with a weighted blunt four-prong grapnel, CO2 soda-bottle propulsion, NiMH rewind with capacitor-burst assist, and a replaceable cassette — whipper-snipper lineage, his own words' family. What it is for: breach and drift events, where the operator notation is seated verbatim — a hull breach does not suck, it blows, briefly, and then the room is just a room; the gauntlet gives the crew member a thrown anchor, not a movie rescue. The audit's own note rides in §1: the precedents are older than the joke. The vet graded the tether mechanism a pass; the remaining tests are ordinary — line strength, grapnel retention, CO2 pressure, launch velocity, recoil, rewind torque, cassette swap, snag behavior — listed in §9.

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat (the audit). Chapter GP-IM-165, v1.0 — 23 August 2026.

§0 — READ THIS FIRST (HOW TO USE THIS CHAPTER)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

Section map: §0 this page — §1 the gauntlet law as seated (contents line, the bodies — the device, what it is for, the Hollywood-suction operator notation, the physics note — verbatim) — §2 the throw graded — §3 the system on the wrist — §4 the doctrine record — §6 build sequence — §7 cross-phase — §8 do-not-build — §9 owed — §10 status, provenance, change control.

§1 — THE LAW (AS SEATED, VERBATIM)

THE CONTENTS LINE (verbatim): 'Phase 165: **Tether Gauntlets — Wrist-Mounted Line Throwers.** Short arm gauntlets firing thin Kevlar line with a weighted blunt four-prong grapnel; CO2 soda-bottle propulsion, NiMH rewind with capacitor-burst assist, whipper-snipper cassette spools. Lifelines for violent hull events, meteor showers, and planetary work where a branch is a tool.'

THE CORE OBJECTIVE: none seated as a standalone line — the contents line above carries the register wording.

Live instruction, 29 July 2026. Written in real time with Archer, immediately after the Phase 163 lone-casualty extension. This phase is the first general-utility design of the post-161 era — born from the same question as the bulkhead wall stations (what does one crew member do when help isn't coming?) and answered with a device worn, not stored.

THE DEVICE (VERBATIM)

- **The Gauntlet:** arm bracelets — short gauntlets — racked on the wall of the bridge, every personal pod, and every work pod. Worn on the forearm, fired from the wrist.
- **The Line:** thin Kevlar thread — enormous tensile strength for near-zero carried mass. The end is weighted: a non-sharp, four-pronged climbing harpoon that catches structure without piercing hull, flesh, or suit.
- **The Propulsion:** CO2 gas models using small soda-bottle-sized units — proven pressure in a replaceable cartridge.
- **The Recovery:** NiMH rewind pack with capacitor-burst assist — battery-powered line recovery per the Phase 173 fleet battery chemistry mandate — plus a wind-up handle function, or a whipper-snipper cord-style replacement cassette when the spool is spent.

WHAT IT IS FOR (VERBATIM)

- **Violent Events:** meteor shower, attack, hull breach — when the ship moves violently, fire a lifeline before the deck moves out from under you. Aid yourself, then aid others.
- **Rescue:** reaching a trapped person after an event — a line thrown across a wrecked section beats a body climbing through it.
- **Logistics:** pulling supplies through a narrow aperture — the line goes where the crew cannot.
- **Planetary Exploration:** pulling a branch down for fruit or seed — you are not carrying a ladder around an alien world. Our AI will not be TV-superhuman in strength; the tether multiplies what limited actuators and ordinary arms can reach and move.

OPERATOR NOTATION — THE HOLLYWOOD SUCTION MYTH: WHAT A BREACH ACTUALLY DOES (30 JULY 2026) (VERBATIM)

- **14 psi is not a bomb.** One atmosphere is the pressure every crew member has lived in since birth. A hole does not turn it into a bomb; it turns it into a wind — and a wind bounded by the hole's own area. The pressure difference can only push on what sits directly in front of the opening. It reaches

nobody across the room.

- **Air escapes by volume-to-hole — and then it stops.** Large hole: the compartment's air is gone in seconds — instant equalization, and after that there is no flow at all. Small hole: a local draft that starts at 14 psi and declines steadily to zero as the compartment empties. There is no sustained movie-length gale; the physics has a closing time.
- There is no venturi effect in space. An aircraft fuselage breach has dual physics — cabin pressure pushing out plus a fast slipstream actively sucking air off the skin. A hull in vacuum has no slipstream. Proof by observation, in Archer's framing: the maintenance crew member standing on the outside of the ship, working in hard vacuum, goes nowhere. There is no wind out there to move anyone.
- **Nobody dangles on a tether being 'sucked' toward a hole for an extended period.** The outflow window near a large opening is real but measured in seconds — and that window is exactly why the gauntlet exists: fire the line, arrest the drift, and by the time a film would still be screaming, the air is gone and so is the force. What remains is a compartment at zero pressure and a person on a line, in charge of their own movement.
- **Breathing is the problem — not horizontal holding-on-to-a-plane-wing physics.** Sudden vacuum allows roughly 9-15 seconds of useful consciousness. That is why Phase 163 stations pair every extinguisher and repair gun with handheld oxygen breathers: the casualty model is suffocation on a clock, never suction.
- **The tether's true job is controlled movement in a zero-G vacuum.** For a crew that lives in gravity, a breach event's real shock is disorientation — the deck stops mattering. If you are already in a float environment, only movement alters. And a gravity ship does not lose its gravity: spin and thrust do not leak through a hole, vacuum or otherwise.
- **Audit confirmation (Kimi):** the note is dimensionally correct. Flow through the aperture is choked at the speed of sound at the throat, but room-side velocity falls off with distance squared from the hole — a body even a metre away from a small breach feels a draught, not a tow. Equalization time scales with compartment volume divided by hole area: seconds for a large strike, minutes of weakening draft for a puncture. The 9-15 second useful-consciousness figure is the human-factors standard for one-atmosphere sudden decompression.

Archer, 30 July 2026: "people being sucked into space in movies is pretty much bullshit. 14psi is not a bomb and air escapes by volume to the hole, the brain rectifies this quickly that this is not a planes that has the dual physics off pressure balance and the venturi effect, there is no venturi effect in space, and there is a guy standing on the outside of the ship doing maintenance going nowhere so there is no one hanging on the end of a tether with their body being sucked towards a hole for some movie length extended period, its instant with a large hole and the pressure from a small hole moving air is only 14psi down to zero. breathing is the problem not horizontal holding on to a plane wing physics. the tethers are

for safe controlled movement in the now zero g vacuum, more of a shock to a crew living in a gravity environment. if you are already in a float environment only movement alters and realistically a gravity ship would not lose its gravity vacuum or otherwise. i love star trek the movies, but is like having the hiccups say bullshit every minute or so"

PHYSICS NOTE — PHASE 165 (KIMI AUDIT: AFFIRMED — THE PRECEDENTS ARE OLDER THAN THE JOKE) (VERBATIM)

- **"As silly as it may sound" — it is not silly, it is SOLAS.** Ocean ships have carried line-throwing apparatus for over a century: a projectile carrying a light messenger line across 230+ metres, fired to another vessel or a wreck, so the heavy line can follow. It is mandated rescue equipment under international maritime law. The gauntlet is the same device at wrist scale, and the Kevlar messenger line plays exactly the messenger's role — light line first, load after.
- **Kevlar is the right cord.** Aramid fiber runs roughly five times the tensile strength of steel at equal weight; a thread-scale line still holds a crew member's mass with orders of magnitude to spare, and its near-zero mass is what lets a soda-bottle CO2 charge throw it usefully far. The heavy, blunt, four-pronged head is correct grapnel doctrine: mass to carry the line, prongs to catch structure, no edge to damage hull lining, suits, or the person it is meant to save.
- **CO2 soda-bottle propulsion is proven at exactly this scale.** Paintball markers, air pistols, and soda systems all run on the same liquid-CO2 cartridge physics — roughly 60 bar of self-pressurizing gas, no pump, no compressor, shelf-stable for years. The whipper-snipper cassette is equally proven consumer engineering: a pre-wound spool swapped in seconds, which matters because a line thrown into a wrecked corridor may not come back.
- In zero-G the tether graduates from safety gear to transport. Astronauts on ISS EVA already live by the tether rule; a fired line plus a rewind winch is personal propulsion — pull yourself to the anchor, or pull the anchor's cargo to you. One caution for the training chapter: firing throws mass forward, so Newton throws the shooter backward — brace against structure before firing, especially in microgravity, and never fire from a floating stance.
- **The honesty about AI strength is the design's spine.** Actuators will be sized for mass and power budgets, not heroics — so the design hands reach and leverage to the machine instead of demanding strength from it. A line thrown by a modest arm, anchored by a grapnel, rewound by a winch, converts a strength problem into an energy problem, and energy is the one thing a starship always has.

ORIGIN OF THOUGHT — PHASE 165 (VERBATIM)

Archer, 29 July 2026: "the bridge and all personal pods and work pods will have, arm bracelets like short

gauntlets on the wall. as silly as it may sound these fire tethers like spiderman, just thin Kevlar thread the end is heavy with non sharp 4 pronged climbing harpoon, lithium rewind ability. this is for an incoming meteor shower attack etc for violent movement and hull breach's to fire a lifeline to aid yourself or others. it may not be anything other than reaching a trapped person after an event, pulling supplies through a narrow aperture. our AI wont be tv superhuman in strength, on planet exploring to pull a branch down for fruit or seed, you aint carrying around a ladder etc etc. co2 gas models using small soda bottle sized units would work with a wind up handle function or a whipper sniper cord style replacement cassette."

Amendment, 30 July 2026 — Archer: "NiMH for the gauntlets (my fault is say lithium because its generic for small high power)." The rewind pack is NiMH with capacitor-burst assist per the Phase 173 fleet battery chemistry mandate; the spec bullet in this phase carries the correction. The 29 July quote above stays verbatim.

Why it matters, in plain English: every emergency design until now waited at a fixed point — the pod, the wall station, the kit. The gauntlet is the first one that walks out the door with you. It answers the hole left in Phase 163's lone-casualty rule: once the casualty is safe, the next problem is distance — the wreck between you and the trapped crew member, the aperture too small for a body, the deck that won't stay still during a breach. And it is honest about the machines: the fleet's AI will be built to mass and power budgets, not comic-book strength, so the design gives them reach and leverage instead. A device that rescues a crew member in a meteor shower and pulls fruit off a branch on a new world is not two devices — it is one idea, honestly scaled: light line first, load after.

§2 — WHY IT WORKS (THE PHYSICS, GRADED)

THE THROW, AFFIRMED: CO2 propulsion for a light line is correct sizing — a soda-bottle charge throws metres of Kevlar without a compressor, and the capacitor-burst rewind gives fast recovery without a heavy motor. The vet's own check ran in the file: firing throws mass forward, so the shooter receives backward momentum — in microgravity, brace before firing; the line is not reactionless propulsion.

THE GRAPNEL, AFFIRMED: weighted, blunt, four-prong — it wraps and hooks rather than penetrates, which is what you want near a pressure hull and a crew.

THE OPERATOR NOTATION, SEATED (his 30 July words, verbatim in §1): the Hollywood suction myth is dead law in this file — a breach equalizes in seconds; after that the gauntlet's job is anchoring and retrieval in a still room. Designing for the real event instead of the movie is the phase's quiet virtue.

§3 — THE SYSTEM ON THE WRIST

- THE GAUNTLET: short arm frame, wrist-mounted.

- THE LINE: thin Kevlar, weighted blunt four-prong grapnel.
- THE CHARGE: CO2 soda-bottle propulsion.
- THE REWIND: NiMH with capacitor-burst assist.
- THE CASSETTE: replaceable — reload in seconds.
- THE BRACE: the operator rule — in microgravity, brace before firing (the vet's check, seated).

§4 — THE DOCTRINE RECORD (HOW THE BOOK ALREADY RUNS ON IT)

- THE SPIDER-MAN RULE (the phase's own name): everyone carries the line — personal, not locker-stowed.
- THE REAL-EVENT DOCTRINE: design for the breach that physics delivers, not the one cinema sells.
- THE OLD-PRECEDENT DOCTRINE (the audit's note): line-throwers are older than the joke — maritime line guns, whipper-snipper spools; the file records its ancestors.
- THE MOMENTUM RULE: every throw is Newton's third law — brace, then fire.

§5 — VALIDATION: the grade and its full record live in the master and the restart.

§6 — BUILD SEQUENCE

- STEP 1 — Build the gauntlet: frame, spool, charge port, rewind drive.
- STEP 2 — Bench the throw: launch velocity and grapnel carry across charge pressures.
- STEP 3 — Bench the line: breaking strength, abrasion, snag behavior.
- STEP 4 — Bench the retention: grapnel set-and-hold across surface types.
- STEP 5 — Bench the rewind: NiMH and capacitor-burst torque, cycle life, cassette swap time.
- STEP 6 — Measure the recoil: operator impulse per throw — the bracing rule quantified.
- STEP 7 — Publish the ledger: all of it — open under the license.

§7 — CROSS-PHASE (INLINED IN FULL)

- PHASE 20 — the tether network: the ship-scale family of the same idea.
- PHASE 188 — the monkey net: spread capture when the target is a person.
- PHASE 163 — the casualty pods (GP-IM-163): drift rescue feeds the triage routing.

- PHASE 166 — the vitals watches (GP-IM-166): the alarm that tells the wrist when to throw.

§8 — DO-NOT-BUILD

- DO NOT fire unbraced in microgravity — the recoil is real and the file has it in writing.
- DO NOT sharpen the grapnel — blunt prongs near a hull; penetration is the rejected failure.
- DO NOT movie the breach — the suction myth is struck; the room equalizes in seconds.
- DO NOT stow it in a locker — the rule is personal carry; a tether in a cabinet saves nobody.

§9 — OWED

OWED: the ordinary bench list: line breaking strength; grapnel retention; CO2 pressure/temperature; launch velocity; recoil; rewind torque; cassette replacement; snag behavior. None is a physics objection. Seat additively as benched.

§10 — STATUS / PROVENANCE / CHANGE CONTROL

STATUS: CHAPTER GP-IM-165, v1.0 — CURRENT, seated law, master v2.1 (numerical print order). Built 23 August 2026, eighteenth of the 20-chapter 'ok next 20, that will take us past gemini' shift (the author's order, 23 August 2026). First version until publication — 'until i publish it, it is still first version, otherwise is just typing' (his law, 22 August 2026).

PROVENANCE — THE STANDING ORDERS, VERBATIM: the town-trip order (seated in sitting 519): 'ok i have to go to town, i have converted them to pdfs now and am uploading, do the next 10 phases and i will read them when i get back' — the order that opened the chapter program; the follow-on orders (seated in sittings 537, 557, 561, 562, 564, 566, 572, 576 and 587): 'ok next 10' / 'ok next ten' / 'all good next 10' / '10 more before go to the old lady for dinner' / 'do another 20 while i away so we clear the 100 mark for the day when i get back' / 'ok next 20' / 'ok next 20, that will take us past gemini'.

CHANGE CONTROL: this design changes only on the Author's orders, seated as sittings in the master record. If the master and this chapter ever disagree, THE MASTER WINS. This is a workshop chapter — it carries the current design only; the full change record lives in the master and the restart.

END OF CHAPTER 165